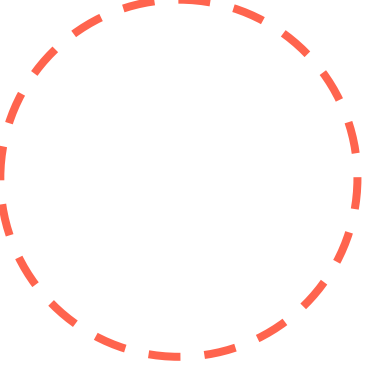




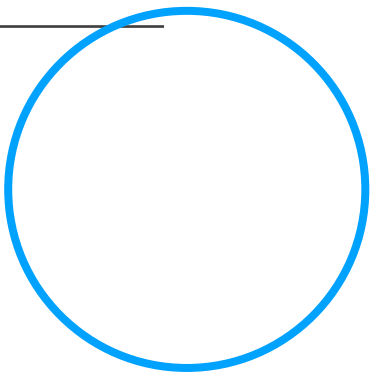
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Based



KNOWLEDGE / SYSTEM

INTRODUCTION (PART 2)



OUTLINE

- ~~Why you need to learn this subject?~~
- ~~What is KBS?~~
- ~~KBS: Early works~~
- ~~DKIW~~
- Types of Knowledge
- Types of Experts
- KBS Structure
- KBS vs Conventional System
- KBS Lifecycle
- KBS Language, Shells & Tools
- KBS Advantage/Limitation



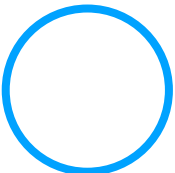
Type of Knowledge

Type of Knowledge



Knowledge Type	Description
Domain knowledge	Domain knowledge is valid knowledge for a specified domain. Specialists and experts develop their own domain knowledge and use it for problem solving.
Meta knowledge	Meta knowledge can be defined as knowledge about knowledge.
Commonsense knowledge	Common sense knowledge is a general purpose knowledge expected to be present in every normal human being. Common-sense ideas tend to relate to events within human experience.
Heuristic knowledge	Heuristic is a specific rule-of-thumb or argument derived from experience.
Explicit knowledge	Explicit knowledge can be easily expressed in words/numbers and shared in the form of data, scientific formulae, product specifications, manuals, and universal principles. It is more formal and systematic.
Tacit knowledge	Tacit knowledge is the knowledge stored in subconscious mind of experts and not easy to document. It is highly personal and hard to formalize, and hence difficult to represent formally in system. Subjective insights, intuitions, emotions, mental models, values and actions are examples of tacit knowledge.

Table 1: Types of Knowledge



Type of Expertise

- An expert is a person who possess many different levels of skills or knowledge.

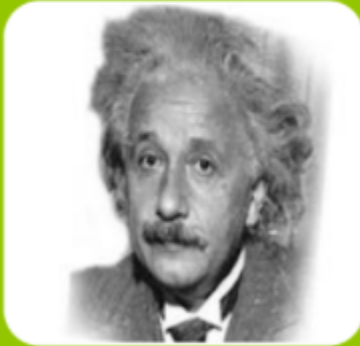


Motor Skill

- Physical rather than cognitive-oriented; therefore KBS cannot easily emulate this type of expertise.
- Humans learn these skills by **repeatedly performing** them.
- **E.g.,**
 - Riding a bicycle
 - Kicking a football
 - Wakeboarding or water skiing

Type of Expertise

- An expert is a person who possess many different levels of skills or knowledge.



Theoretical (Deep) Knowledge

- Finding solutions to technical problems often requires going beyond a superficial understanding of the domain.
- Must apply creative ingenuity.
- Based on **theoretical knowledge** of the domain.
- Cannot be solved via associational expertise.
- Acquired via **formal training and hands-on problem solving**.
- Very easily forgotten unless continually used, due to theoretical and often abstract nature.
- E.g., engineers and scientists.

Type of Expertise

- An expert is a person who possess many different levels of skills or knowledge.

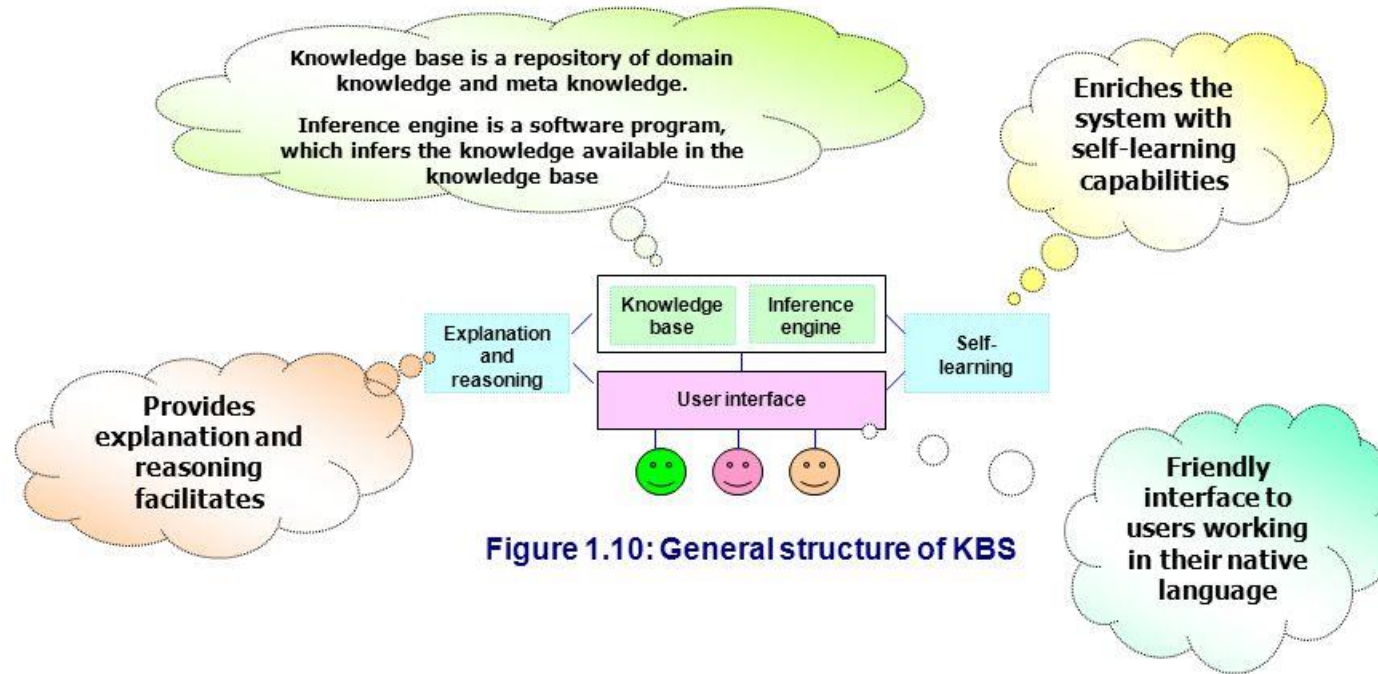
Associational Knowledge



- KBS excel in representing associational knowledge.
- This level of skill reflects **heuristic ability or knowledge** that is acquired mostly through observation.
- The expert **may not understand what happens inside a black box**, but he can associate the inputs with specific outputs.
- **E.g., TV repairman**
 - can fix nearly all common problems but does not understand transistor theory or CRT/LCD/plasma display theory
 - based on experience, rather than analysis

KBS General Structure

Components of KBS





- 

- mechanism **to derive new knowledge** from the knowledge base and the information provided by the user
- often based on the use of rules



- 



Conventional System vs KBS



Conventional

Info and processing
combined in 1
sequential program

The program does not
make mistake

Do not explain why

Require all input data

Changes in program
are tedious

Expert System

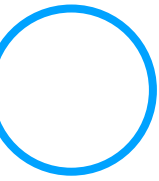
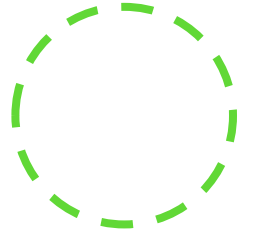
Knowledge is
separated from the
processing (inference)

Program makes
mistakes

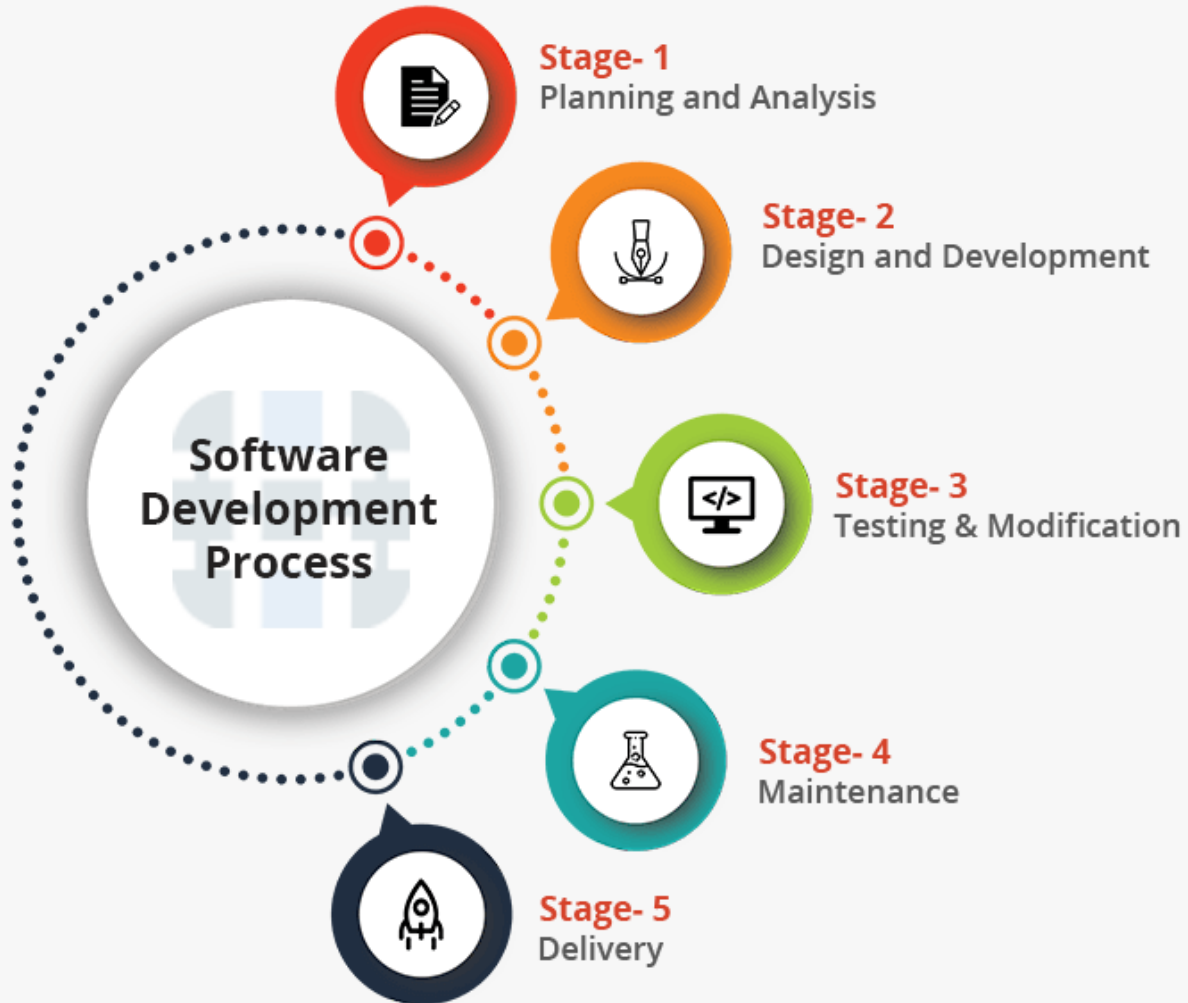
Explanation is part of
the system

ES do not require all
initial facts

Changes in rules are
easy to make



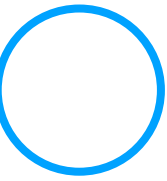
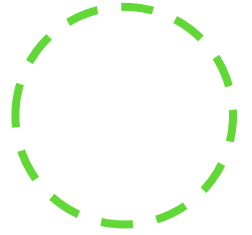
General Software Development Process



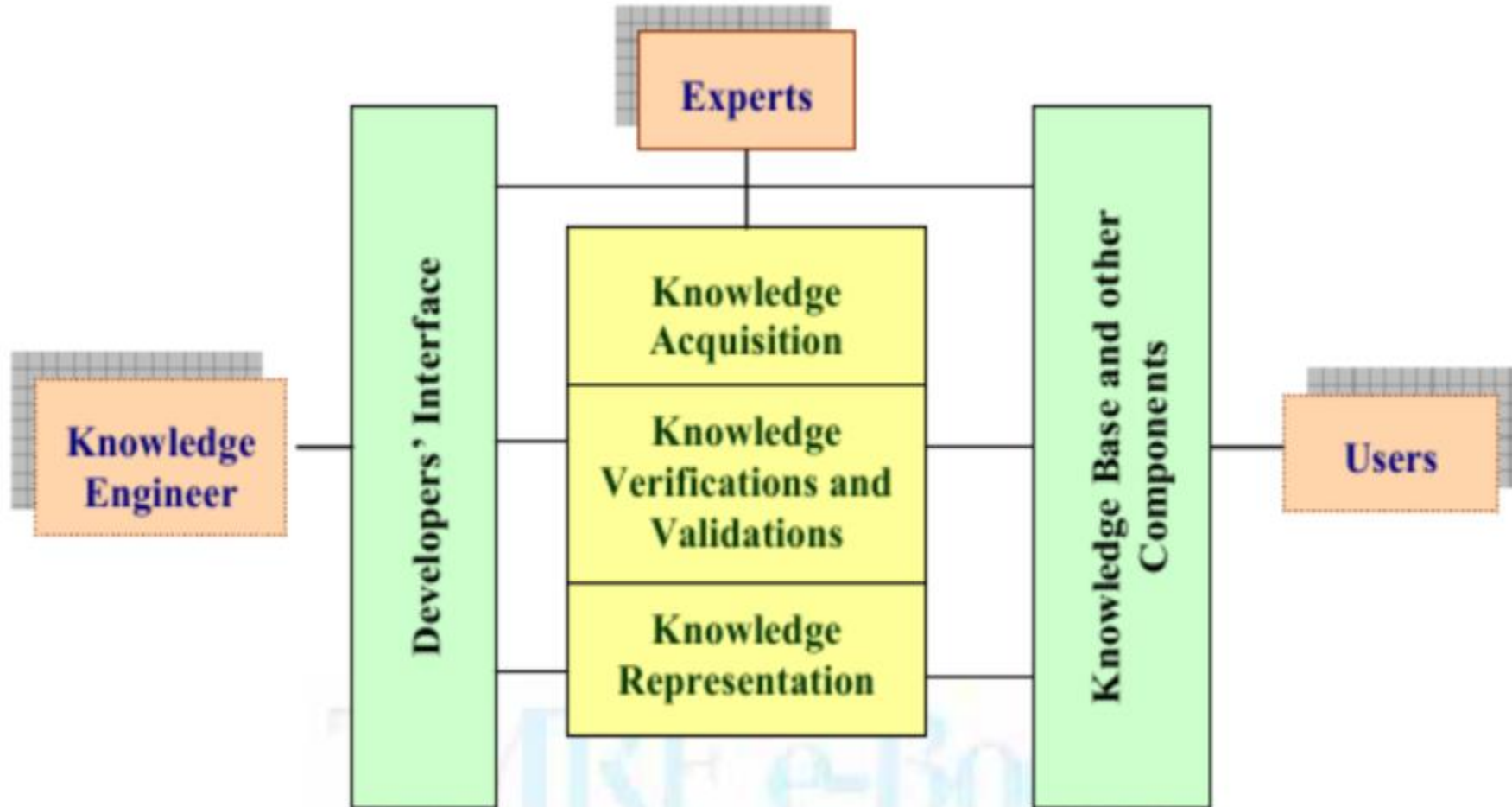


KBS **Development** **Process**

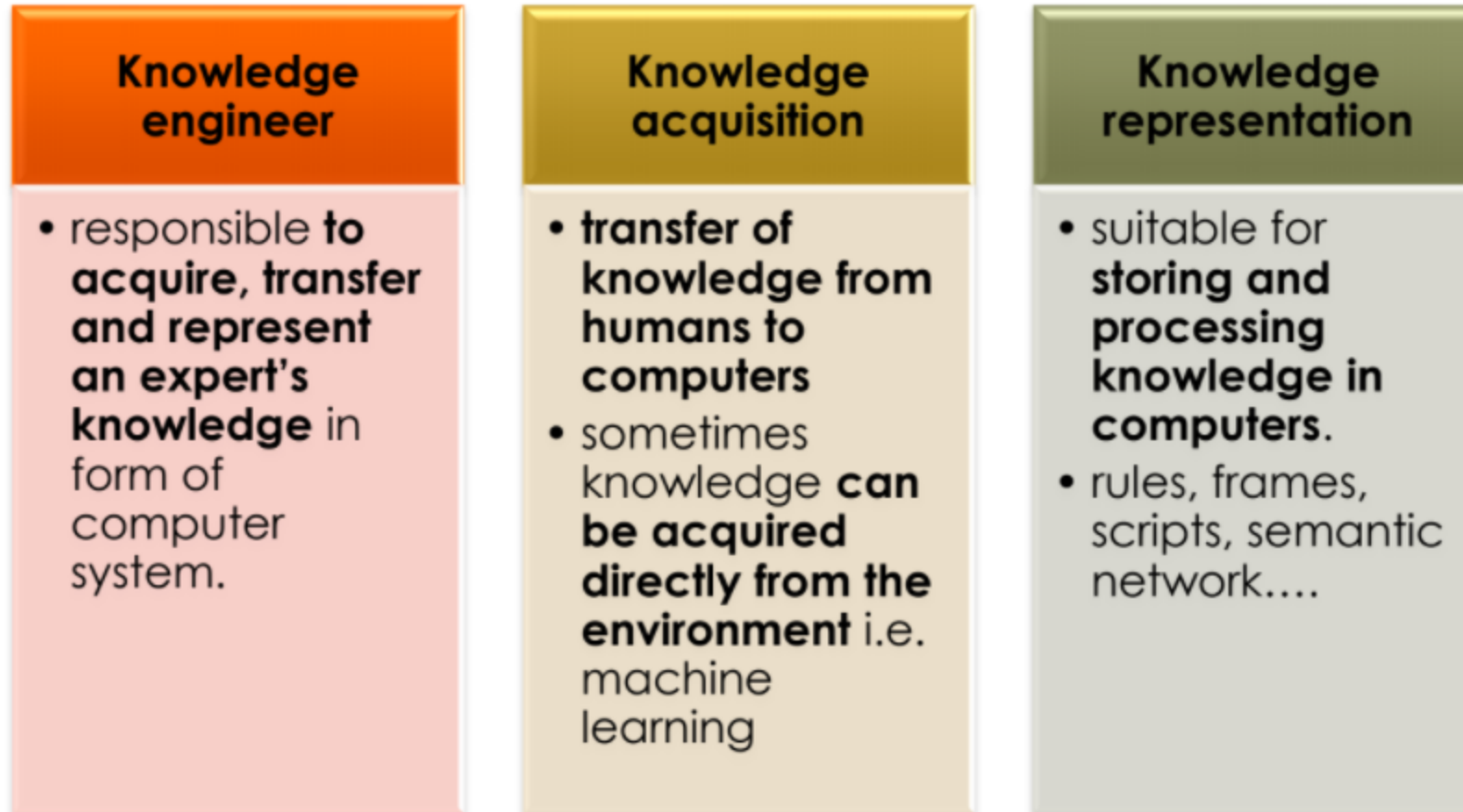
Planning	
Knowledge Formulation	
Source ID & Selection	Acquisition Analysis & Extraction
Knowledge Base Design	
Formulation	Detailed Design
Programming & Debugging	
Knowledge Verification with Industrial Experts	
Formal Test	Test Analysis
System Evaluation	



KBS Development Process



KBS Development Process

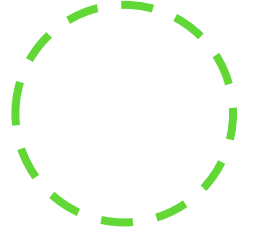


Languages, Shells, and Tools

- Expert system languages are post-third generation.
- Expert system languages (e.g. CLIPS) focus on ways to represent knowledge.
- Tool = language + utility program (code generator, graphics editor, debuggers, etc.).
- Shell: is a special purpose tool designed for certain types of applications in which the user must supply the knowledge base. Example, EMYCIN (empty MYCIN)

Tool

EMYCIN (Lisp) Stanford
EXPERT (Fortran) Rutgers
KEE (Lisp) Intellicorp
VP-Expert
PC-Plus (Lisp) Texas
Instruments
GoldWorks (Lisp) Bolesian
Systems
ART (Lisp)
EXSYS
APEX Fault diagnosis shell
NEXPERT (C) Neuron Data
KAPPA Intellicorp
DELFI (TurboPascal) Delft
University
Pro.M.D. (Prolog)



Benefits and Limitations



When **expert is not available**.

When expertise is to be **stored for future use**.

When intelligent assistance and training are required **for decision making**.

When **more than one experts' knowledge have to be grouped** at one platform.

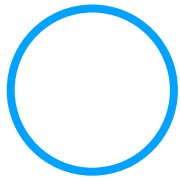


Acquisition, representation and manipulation of the **large volume of the data** /info /knowledge.

High-tech image of the AI field.

Abstract nature of the knowledge.

Limitations of cognitive science and other scientific methods.



Benefits and Limitations



Wide distribution of scarce expertise.

Ease of modification.

Consistency of answers.

Perpetual **accessibility**.

Preservation of expertise.

Solution of problems involving **incomplete data**.

Explanation of solution.

Answers may **not always be correct**.

Knowledge **limited to the domain of expertise**.

Lack of common sense.

